



Full length article

A multi-modal and multi-stage fusion enhancement network for segmentation based on OCT and OCTA images

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ARTICLE INFO

Keywords:

Optical coherence tomography
Optical coherence tomography angiograph
Feature fusion
Multi-modal

ABSTRACT

The accurate segmentation of retinal vessels (RV) and Foveal Avascular Zone (FAZ) is crucial for the retina health assessment. However, fusing OCT and OCTA images while projecting these 3D data for segmentation is a challenge. In this paper, we propose an Adaptive Projection and Multi-stage Fusion Enhancement Network for the segmentation of RV and FAZ, which consists of 3D projection, fusion enhancement, and 2D segmentation parts. We design a kernel adaptive unidirectional convolution to improve projection efficiency and a volume fusion module to capture semantic and spatial information accurately. Then we construct a Multi-stage Complementary Fusion Module and a Multi-modal Difference Fusion Module to enhance features by complementarity and differences in features. At last, we employ three U-Nets with a Feature Interaction Module for segmentation. Experiments show that our method achieves state-of-the-art performance with DICE, BACC, and JAC of 0.8996, 0.8181, 0.9421, 0.9395, 0.8923, and 0.9717 for RV and FAZ.

1. Introduction

Retinal vessels (RV), as the only deep microvessel in the human body that can be non-invasively observed, play a vital role in diagnosing various eye-related diseases such as hypertension, diabetic retinopathy, and arteriosclerosis [1]. RV map incorporate a wealth of geometric features like vessel diameters, branching angles, and branching lengths. These geometric characteristics reflect clinical and pathological traits, serving as critical indicators for the detection and diagnosis of these diseases [2]. The Foveal Avascular Zone (FAZ) is typically located at the center of the macula, lacking capillaries. The shape and size of the FAZ are closely related to age, visual acuity, and various ocular diseases [3]. For instance, an enlarged FAZ is usually associated with impaired vision [4], while an irregular FAZ is linked with age-related macular degeneration [5,6] and glaucoma [7]. Moreover, the curvature diameter of the FAZ can reflect the condition of diabetic retinopathy to some extent [8,9], which makes FAZ as a crucial indicator in clinical practice. Therefore, the research on vascular and FAZ segmentation techniques based on retinal images has important clinical and practical value for auxiliary diagnosis and treatment of various diseases related to it.

Optical coherence tomography (OCT) is a non-contact imaging technique which generates cross-sectional images of tissue with high resolution [10]. While optical coherence tomography angiography (OCTA) is an emerging type of OCT functional imaging [11], as shown in Fig. 1.

Compared to OCT, OCTA can provide functional information on the retinal vascular and microvascular systems. The enormous advantage of OCTA technology in providing real-time cross-sectional images of tissue structures with blood flow information at the micron level has made OCTA an important imaging tool for clinical diagnosis of retinal diseases, which has got widespread attention from researchers and has great research significance.

However, the majority of current RV segmentation primarily adopt fundus images as data sources, rather than OCT. Compared with fundus images, OCT provides more detailed information, including retinal layer structure, lesion morphology and extent, etc, but its ability to display vessels information is limited. The emergence of OCTA has sparked researchers' interest in developing RV segmentation based on the promising data. For FAZ, there was limited research about it due to the challenge in quantization. However, the non-invasive nature and high-resolution characteristics of Optical Coherence Tomography (OCT) and OCT Angiography (OCTA) have made the visualization of FAZ clearer, reducing the difficulty of FAZ quantification [12]. Furthermore, the discovery of the FAZ geometric features related to age, vision, and disease has aroused considerable interest among researchers, accelerating the research progress of FAZ segmentation. Because OCT and OCTA data are 3D volume data, and most retinal indicators are quantified on 2D projection maps, it is necessary to convert the 3D data into 2D projection images, as shown in Fig. 1.

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<https://doi.org/10.1016/j.inffus.2024.102594>

Received 12 April 2024; Received in revised form 8 July 2024; Accepted 22 July 2024

Available online 25 July 2024

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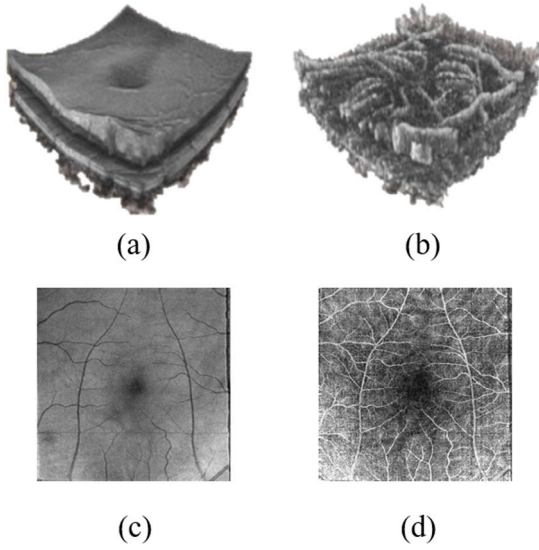


Fig. 1. (a) OCT volume. (b) OCTA volume. (c) projection map of OCT. (d) projection map of OCTA.

The traditional method of retinal blood vessel segmentation for OCTA volume is obtaining 2D projection images through retinal layer segmentation algorithm, and then segment them based on the projection images [13–15]. Recently, Li et al. proposed a novel convolutional neural network-based projection method, called the Image Projection Network (IPN). IPN can compress data along the projection direction in multiple stages through Projection learning module (PLM) composed of 3D convolution and unidirectional pooling, and finally generate a 2D feature map for 2D semantic segmentation [16]. Subsequently, Li et al. proposed IPNv2 on the basis of IPN, which improved the Projection learning module (PLM) to Fast Projection Module (FPM) [17]. FPM composed of unidirectional convolution and unidirectional pooling has downsampling operations in the horizontal direction and smaller memory footprint compared with PLM.

The introduction of IPN and IPNv2 paved the way for exploring OCT and OCTA. However, there is a problem in IPN-V2 where the differences between the OCT modality and OCTA modality are not taken into account. They only use OCT or OCTA data as inputs, and the results of using OCT and OCTA as inputs simultaneously are inferior to using OCTA alone, which indicates the presence of modality imbalance problem. Therefore, the paper proposes a new network architecture called Adaptive Projection and Multi-stage Fusion Enhancement Network (APMFENet), which enhances the projection effect through the multi-stage fusion and utilizes attention mechanisms to fuse multi-modal and multi-stage features, building interpretable models to address the modality imbalance problem between OCT and OCTA. APMFENet consists of three main components: the 3D projection part, the fusion enhancement part, and the 2D segmentation part.

In the 3D projection part, we introduce a Kernel Adaptive Projection Module (KAPM) to improve the compression efficiency of 3D data. The module proposes kernel adaptive unidirectional convolution, which selects different convolutional kernels to project different parts of the same volume data. At each projection stage, we use the multi-stage mid fusion method and replace the simple summation in multi-stage mid fusion with a Volume Fusion Module (VFM), which can fuse features at different stages of the network and then share half of the backbones to extract the fused features. The method accomplishes data reuse and can accurately capture shallow spatial information and deep semantic information.

In the fusion enhancement part, we introduce the Multi-stage Complementary Fusion Module (MCFM) and Multi-modal Differential Fusion Module (MDFM), which further fuse features by the inherent

complementarity and differences in features, and alleviate the issues of modality imbalance caused by misalignment and insufficient fusion of different modalities.

Finally, in the 2D segmentation part, we designed a Feature Interaction Module, which is used to prevent uneven feature contribution.

In summary, this paper has the following contributions:

(1). We propose a novel framework called the Adaptive Projection and Multi-stage Fusion Enhancement Network (APMFENet), designed to model OCT and OCTA volumetric data for the joint segmentation of FAZ and RV.

(2). In the 3D projection part, we introduce a Kernel Adaptive Projection Module (KAPM) that employs convolutional kernels of different sizes based on the spatial differences in the data, which enhance the projection ability significantly. Furthermore, we utilize a multi-stage mid fusion method with Volume Fusion Module (VFM) to capture shallow spatial and deep semantic information, which endows our model with superior generalization performance.

(3). In the fusion enhancement section, a Multi-stage Complementary Fusion Module (MCFM) and a Multi-modal Differential Fusion Module (MDFM) were designed based on more blood flow information in OCTA and clearer FAZ in OCT. MCFM utilizes the complementarity between OCT and OCTA data, while MDFM extracts features based on the differences between OCT and OCTA data, thereby reducing the modality differences and alleviating the modality imbalance problem.

(4). In the two-dimensional segmentation section, we design three UNet structures and a feature interaction module, which regards the results and their sum of MCFM and MDFM as three inputs, and assign them different loss function coefficients to guide their balance optimization.

(5). We conducted extensive experiments, which fully validate the effectiveness of the network. The experimental results showed that APMFENet outperformed other state-of-the-art FAZ and RV segmentation methods on the OCTA_6M and OCTA_3M parts of the OCTA-500 dataset.

The rest of this paper is structured as follows. In Section 2, we review the related work. Section 3 introduces our method. In Section 4, we introduce different experimental designs and discuss the results. Finally, the proposed work is concluded in Section 5.

2. Related works

2.1. Retinal vessel and foveal avascular zone segmentation

Recently, due to significant advancements in deep learning technology for image processing, many deep learning methods are widely used in the field of retinal vascular segmentation.

The deep learning algorithms for fundus image segmentation are mainly divided into three categories: CNN based algorithms, Transformer based algorithms, and algorithms combined by CNN and Transformer [18]. U-Net is one popular algorithm based on CNN since its efficient feature extraction capability for medical datasets and have derived many improved versions [19–21]. U-Net extracts deep and shallow features through encoder downsampling, and then decodes a larger feature map containing deep and shallow features by the decoder. Skip connection achieve a retrieval of edge features which was lost during downsampling. Medical-Transformer is a medical segmentation model based on gated axial attention mechanism [22]. The axial-attention mechanism is a kind of self-attention which is calculated on two different coordinate axes, and adds a learnable position encoding to obtain good non local information [23]. However, for low scale medical images, the added position encoding may cause significant errors and make it difficult to learn accurate position information. Therefore, a gated unit has been added to reduce the weight of the position encoding. There are also algorithms based on the combination of CNN and Transformer, such as TransUNet [24]. It introduces a Transformer for the U-Net encoder, which has the powerful ability of

extracting detailed information from CNN while also capturing long-distance dependencies, achieving hybrid encoding between CNN and Transformer. Although these algorithms are not applicable to OCT and OCTA volume data, they still provide valuable insights and point the way forward. Consequently, a series of segmentation algorithms for OCT and OCTA data have emerged. Li et al. proposed an end-to-end image projection network (IPN), which adopts unidirectional pooling to achieve 3D to 2D image segmentation in OCTA and OCT images [16]. Additionally, they proposed an IPNV2 network based on IPN, which supplements horizontal downsampling operations, incorporates more advanced semantic information, and achieves excellent results in challenging CAVF tasks [17]. Wu et al. proposed PAENet, which employs an adaptive pooling method to reduce feature dimensionality and capture 3D information for segmentation efficiently [25]. Weisheng Li et al. introduced RPS-Net, which utilizes a bidirectional projection module to learn global perceptual features and local supplementary details, achieving joint segmentation of RV and FAZ tasks [26].

However, these methods did not take into account the difference between the two modalities of OCT and OCTA, and failed to fully exploit the complementarity of information between OCT and OCTA.

2.2. Multi-modal fusion

Multi-modal fusion refers to the integration of data from different modalities and fully utilize the information advantages of each modality. Multi-modal fusion can be divided into early fusion, mid fusion, and late fusion. Early fusion involves simply concatenating multi-modal data before inputting it into the neural network. Mid fusion occurs during the feature extraction process, and late fusion involves fusing output features at the decision layer. Early fusion can effectively extract spatial information, late fusion can accurately capture semantic information, and mid fusion falls between the two [27,28]. Multi-modal fusion algorithms are widely applied in the medical field and hold significant potential for advancements in auxiliary diagnosis and treatment. He et al. proposed a co attention block based cross attention suitable for multimodal tasks, which enhances the representation ability of feature extraction through mutual guidance between two modalities [29].

However, multi-modal image fusion encounters some problems inevitably. Kailai Zhou et al. pointed out the issue of modality imbalance between multi-modal data, which is caused by misalignment and insufficient fusion of different modalities [30]. They further observed that direct concatenation or addition may introduce redundant features, thereby failing to fully exploit the inherent complementarity among features. Weiyao Wang et al. observed that the best single modal networks are usually superior to multi-modal networks, which is primarily attributed to this property that Multi-modal networks are more prone to overfitting, and the overfitting and generalization speeds of different modalities are different [31]. Training multi-modal networks with a single optimization strategy is not suitable and may lead to a modality imbalance problem. To address the problem, Weiyao Wang et al. proposed Gradient-Blending technology to quantify the degree of overfitting and determine the optimal combination of modalities [31].

Additionally, in the field of multimodal fusion for 3D data, Zhu et al. [32] proposed a brain tumor segmentation method based on the fusion of deep semantic and edge information from multimodal MRI, which utilizes a Swin Transformer to extract semantic features, designs an edge detection module based on CNN, and effectively fuses semantic and edge features using graph convolution, thereby achieving feature enhancement. Raza et al. [33] proposed an end-to-end automatic 3D brain tumor segmentation framework. It is a hybrid model of a deep residual network and a U-Net model. The residual network is used as the encoder, while the U-Net model's decoder is employed to address the vanishing gradient problem. Building upon these ideas, we propose a multi-stage mid fusion method, in our research. This approach performs fusion and shares the remaining backbone in multiple stages, enabling the network to extract accurately spatial and

semantic information and achieve volume data reuse. Furthermore, we introduce a Multi-modal Differential Fusion Module (MDFM) and a Multi-stage Complementary Fusion Module (MCFM) to alleviate the issue of modality imbalance.

2.3. Attention mechanism

Attention mechanism, as an effective method for enhancing features, has been applied in various fields of deep learning. HU et al. proposed a Squeeze-and-Excitation Networks (SE-Net) to model the interdependence between channels and enhance the representation ability of CNN [34]. Sang et al. designed a Convolutional Block Attention Module (CBAM), which adds a spatial attention mechanism to SE Net while capturing interdependence in both channel and spatial dimensions [35]. However, CBAM is not able to effectively perform modality fusion tasks. In this paper, we propose a Volume Fusion Module to capture the inter-modal dependencies. Wang et al. proposed a non-local neural network, which capture long-range dependencies in the input data by non-local operation [36]. Ou et al. designed a feature interaction module based on non-local neural network for complementary left and right eye information [37].

In this paper, we propose a Feature Interaction Module based on non-local neural network design to effectively capture global dependencies between features, thereby enhancing the segmentation capability of the network.

3. Methods

3.1. The overall framework

In this section, we introduce present our proposed Adaptive Projection and Multi-stage Fusion Enhancement Network (APMFENet). The framework is shown in Fig. 2, which consists of three parts: the 3D projection part, the fusion enhancement part, and the 2D segmentation part. In the 3D projection part, OCT and OCTA volume data are compressed into 2D through four projections using Kernel Adaptive Projection Module (KAPM). After each projection, a mid fusion operation is performed, which creates a new projection pathway by Volume Fusion Module (VFM), resulting in multiple stages of fused projection maps and unfused multi-modal projection maps. In the fusion enhancement part, the Multi-modal Differential Fusion Module (MDFM) captures unfused multi-modal projection maps, achieving the exchange of complementary information between modalities. Multi-stage Complementary Fusion Module (MCFM) captures the fused multi-stage projection maps and employs spatial attention mechanism to redistribute weights, seeking common representations across different stages. Finally, the two feature maps from MDFM and MCFM along with the projection images are input into the 2D segmentation part.

3.2. Kernel Adaptive Projection Module (KAPM)

To achieve more effectively dimension compression, we propose a kernel-adaptive unidirectional convolution based on the original unidirectional convolution [16], which enhances the compression capability in the vertical direction while preserving the original horizontal directional awareness, as shown in Fig. 3.

In Fig. 3, L, W, H, and C represent the length, width, height, and channel of the volume data. We divide the volume data along the H dimension into K patch of different sizes based on their spatial positions, and adopt convolution operations on each of these small patches. The H dimension of these patches is represented by P. By adjusting the convolution kernel and stride size, the H dimension size after convolution will be 1, then, these K small patches will be concatenated, which is similar to group convolution. But our method divides along the H dimension instead of the C dimension, and each block size is different. At the data edges where the data is sparser,

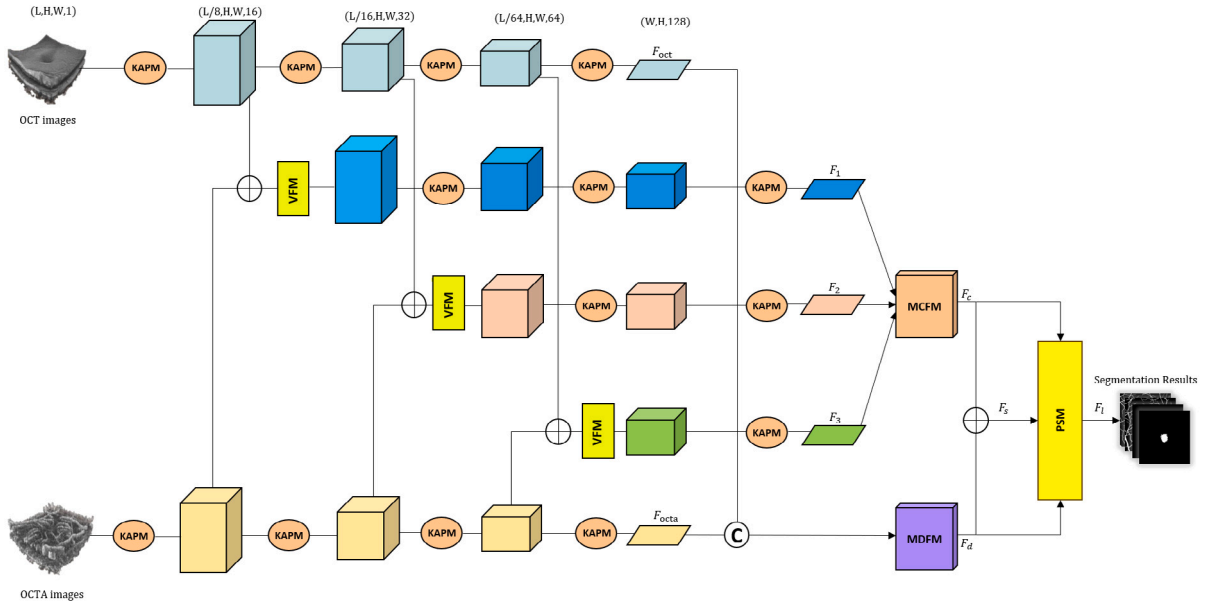


Fig. 2. The overall framework of APMFNet. “ $\oplus$ ” is element-wise addition. “ $\odot$ ” is concatenation.

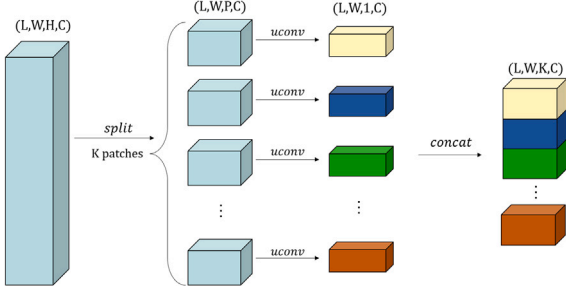


Fig. 3. The structure of kernel-adaptive unidirectional convolution. “uconv” is unidirectional convolution.

we aim to capture more complete edge features and have a larger receptive field. Therefore, we have set slightly larger patches for these regions. Similarly, for the FAZ, which is considered as a whole, we also aim to have a larger receptive field. Hence, larger patches are used in FAZ. For the remaining parts, we expect to extract more detailed features, so smaller patches are employed. The kernel adaptive unidirectional convolution directly compresses the L dimension of each patch to 1 and then concatenates them. The kernel adaptive unidirectional convolution employs different convolutional kernels to compress various positions in space, allowing for more precise parameter selection and enhancing the vertical compression capability. Besides, we design a multi-scale projection network based on the kernel adaptive unidirectional convolution, as shown in Table 1. In KAPM1, KAPM2 and KAPM3, we adopt two scales of convolution for projection. In KAPM4, we employ multiple scales of unidirectional pooling and dilated unidirectional convolution to get features with various receptive fields where kauconv is the kernel adaptive unidirectional convolution, umaxpool is unidirectional maxpooling, k is kernel size, s is stride, and d is dilation rate.

3.3. Volume Fusion Module (VFM)

To better fuse OCT and OCTA volume data, we design a Volume Fusion Module (VFM), as shown in the Fig. 4. The VFM adopts a Squeeze-and-Excitation (SE) block to capture the interdependencies

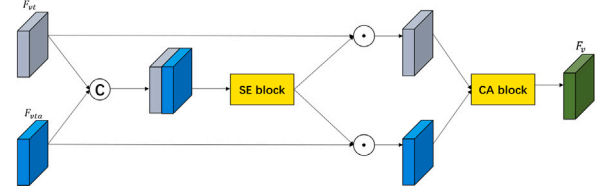


Fig. 4. The structure of Volume Fusion Module. “ $\odot$ ” is element-wise multiplication.

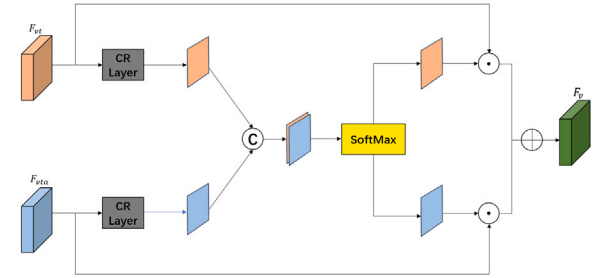


Fig. 5. The structure of SA block.

between OCT and OCTA volume data in channel dimension. Subsequently, a Spatial Attention (SA) block is employed to model semantic dependencies in the spatial dimensions. (i) The SE block consists of global average pooling (GAP), fully-connected layers and a sigmoid activation function. The SE block captures fused features and utilizes GAP to obtain features with a global receptive field. Next, two fully connected layers and a sigmoid function are employed to obtain channel weights that are assigned to OCT and OCTA data. (ii) The structure of SA block is shown in Fig. 5. The SA block utilizes the Convolution ReLU (CR) layers to reduce the number of channels to 1. The obtained features are concatenated and use a softmax function to obtain spatial weights, which are then assigned to OCT and OCTA data where the CR layer consists of two consecutive convolutions, with a ReLU function inserted after the first convolution.

Table 1
The structure of multi-scales projection network.

Stage	Operation	Input size	Output size
KAPM1	$kauconv(k = (a,1,1), s = (a,1,1))$ $kauconv(k = (a,3,3), s = (a,1,1))$	(128,W,H,1)	(16,W,H,16)
KAPM2	$kauconv(k = (a,1,1), s = (a,1,1))$ $kauconv(k = (a,3,3), s = (a,1,1))$	(16,W,H,16)	(4,W,H,32)
KAPM3	$kauconv(k = (a,1,1), s = (a,1,1))$ $kauconv(k = (a,3,3), s = (a,1,1))$	(4,W,H,32)	(2,W,H,64)
KAPM4	$umaxpool(k = (2,1,1/2,3,3/2,5,5), s = (2,1,1))$ $kauconv(k = (2,3,3), s = (2,1,1), d = (1,2,2/1,4,4/1,8,8))$ $kauconv(k = (2,1,1), s = (2,1,1), d = (1,2,2/1,4,4/1,8,8))$	(2,W,H,64)	(W,H,128)

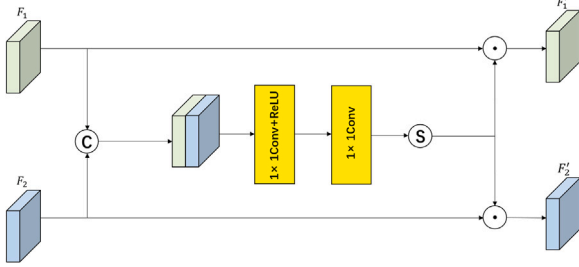


Fig. 6. The structure of Multi-stage Complementary Fusion Module. “S” is sigmoid activation function.

3.4. Multi-stage Complementary Fusion Module (MCFM)

Multi-modal fusion can be divided into early fusion, mid fusion, and late fusion. Early fusion can effectively extract spatial information, late fusion can accurately capture semantic information, and mid fusion falls between the two. Therefore, we designed a Multi stage Complementary Fusion Module (MCFM) that utilizes spatial attention mechanism to capture complementary features at different fusion stages, which can accurately capture of spatial and semantic information. The inputs of MCFM are F_1 , F_2 , F_3 , as shown in Fig. 2. Its specific structure is shown in Fig. 6. Fig. 6 only illustrates the feature fusion of F_1 and F_2 , while the fusion of F_1 and F_3 , F_2 and F_3 are similar to F_1 and F_2 , as shown in Eq. (1) to Eq. (10). We sum up F_1 , F_2 , F_3 to generate feature F_4 , and concatenate F_1 , F_2 and F_1 , F_3 and F_2 , F_3 with each other. Then, we obtain complementary weights through the fc layer and a Sigmoid function. The weight formula is as follows:

$$F_4 = F_1 + F_2 + F_3 \quad (1)$$

$$W_1 = \text{Sigmoid}(fc(\text{concat}(F_1, F_2))) \quad (2)$$

$$W_2 = \text{Sigmoid}(fc(\text{concat}(F_1, F_3))) \quad (3)$$

$$W_3 = \text{Sigmoid}(fc(\text{concat}(F_2, F_3))) \quad (4)$$

$$W_4 = \text{Sigmoid}(fc(F_4)), \quad (5)$$

where fc represents two 1×1 convolutions, and a ReLU function is inserted after the first convolution operation. Subsequently, we multiply these features with the relevant attention weight maps to obtain the enhanced features. Finally, the first three features are summed and concatenate with F_4 to obtain F_c . The formula is as follows:

$$F'_1 = F_1 * (W_1 + W_2) \quad (6)$$

$$F'_2 = F_2 * (W_1 + W_3) \quad (7)$$

$$F'_3 = F_3 * (W_2 + W_3) \quad (8)$$

$$F'_4 = F_4 * W_4 \quad (9)$$

$$F_c = \text{concat}(F'_1 + F'_2 + F'_3, F'_4) \quad (10)$$

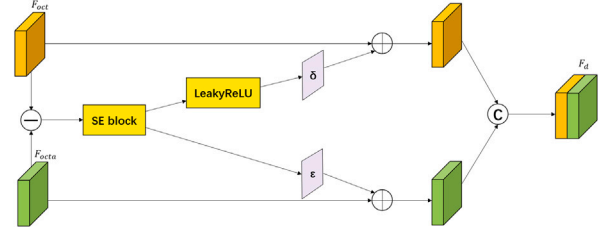


Fig. 7. The structure of Multi-modal Differential Fusion Module. “⊖” is element-wise subtraction.

3.5. Multi-modal Differential Fusion Module (MDFM)

Due to the presence of some irrelevant data for segmentation in OCT and its lack of blood flow information, the fusion of OCT data with OCTA data may result in inferior performance compared with using OCTA alone, which is confirmed subsequent experiments. Therefore, it is necessary to remove redundant data from OCT to enhance them. So we propose Multi-modal Difference Fusion Module (MDFM), as shown in Fig. 7. Inspired by differential amplifiers, the common-mode signal is suppressed, and the differential-mode signal is amplified. Since OCTA data contains blood flow information that OCT lacks, we subtract OCT data from OCTA data to obtain the differential-mode features of blood flow information. We then apply a SE block to capture the inter-channel dependencies. The differential-mode features reflect the information present in OCTA data but absent in OCT data. These differential-mode features are added to the original OCT and OCTA feature map through the gating unit δ and ϵ to further enhance the blood flow features of OCT and OCTA data. Additionally, there inevitably exist some redundant features in the differential-mode features originated from OCT data, which are expressed as negative values. To alleviate the impact of these redundant features to OCTA, the differential-mode features that are inputted into OCTA data need to be passed through a LeakyReLU activation function beforehand. On the other hand, the features from OCT data are expressed as positive values, the incorporation of differential-mode features helps to counterbalance them to some extent. Therefore, there is no need for LeakyReLU activation function. By specifically processing the differential-mode features, the module achieves cross-modal complementarity between OCT and OCTA, improves the sensitivity of the network to OCT modal features, and alleviates modality imbalance problems. Lastly, the enhanced OCT and OCTA features are concatenated to obtain F_d , which is then added to the feature F_c from MCFM to obtain F_f .

3.6. Projection Segmentation Module (PSM)

In the Projection Segmentation Module (PSM), we explored the impact of different backbone for the network. As shown in the Table 2, the U-Net achieve the best overall performance in CAVF (Capillary, Artery, Vein, and FAZ) task on the OCTA_6M dataset [16]. Therefore, we designed a 2D segmentation module based on U-Net. The input

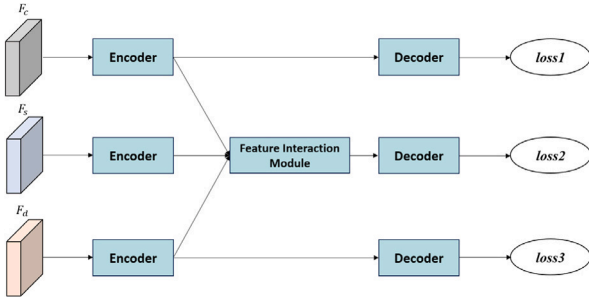


Fig. 8. The structure of Projection Segmentation Module.

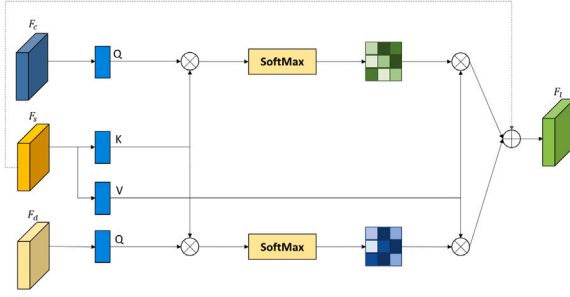


Fig. 9. The structure of Feature Interaction Module.

Table 2
The impact of different backbone for the network.

3D backbone	2D backbone	mIoU
IPN-V2	U-Net	81.37
IPN-V2	Attention U-Net	81.21
IPN-V2	U-Net ++	81.35

of the 2D segmentation module is shown in Fig. 1, which are the results of MCFM F_c , MDFM F_d , and the sum of MCFM and MDFM F_s , respectively, and its specific structure is shown in Figs. 8 and 9. To prevent uneven feature contributions from MCFM and MDFM, we use three U-Nets and Feature Interaction module to segment F_d , F_c , and F_f , and obtain the last result F_l .

4. Experiments

In this section, we will introduce the loss function, dataset, evaluation metrics, implementation details and results of the experiment.

4.1. Loss function

As the most commonly used loss function, cross-entropy can solve majority classification tasks. So we use three cross entropy loss functions, which are specifically described as follows:

$$Loss(x_1, x_2, x_3, label) = \alpha L_{ce}(x_1, label) + \beta L_{ce}(x_2, label) + \gamma L_{ce}(x_3, label) \quad (11)$$

Here x_1 , x_2 , and x_3 represent the segmentation results from the three U-Nets, and label represents groundtruth, α , β , γ represents the weight coefficient. By assigning different-sized loss functions to them, we ensure that the fitting speed of x_1 matches that of x_2 to prevent uneven feature contributions.

4.2. Datasets

We select the OCTA-500 public dataset to evaluate our network. Compared with other OCT and OCTA datasets, OCTA-500 provides

three-dimensional volume data and two-dimensional projection images, which is more valuable for research [38–41]. The dataset is divided into two parts based on field of view (FOV): OCTA_6M for 6 mm × 6 mm FOV and OCTA_3M for 3 mm × 3 mm. The OCTA_6M consists of 300 samples (NO.10001-NO.10300). The size of each sample is 400 px × 400 px × 640 px. To reduce the parameters of our model, we resize the sample size to 400 px × 400 px × 128 px using bilinear interpolation. The OCTA_3M consists of 200 samples (NO.10301-NO.10500). The size of each sample is 304 px × 304 px × 640 px. we resize the sample size to 304 px × 304 px × 128 px using bilinear interpolation. Subsequently, we adopt random horizontal flipping, vertical flipping, and 90-degree rotations as data augmentation techniques to alleviate network overfitting. We divided OCTA_6M into training set (NO.10001-NO.10240), validation set (NO.10241-NO.10250), and test set (NO.100251-NO.10300). and OCTA_3M into training set (NO.10301-NO.10440), validation set (NO.10441-NO.10450), and test set (NO.10451-NO.10500).

4.3. Implementation details

Our network is implemented on the PyTorch framework. All experiments are conducted on a GeForce RTX 3090 Ti GPU. We used the Adaptive Moment Estimation (Adam) optimization algorithm for training, with a first-order momentum of 0.9, second-order momentum of 0.99, and an initial learning rate of 0.003. The total number of iterations for the experiments is set to 300 epochs.

4.4. Evaluation metrics

In order to evaluate the segmentation performance of our network more comprehensively, during the ablation experiment stage, we adopt the challenging CAVF multi-classification task to demonstrate the improvement in overall performance. In this case, we use mean Intersection over Union (mIoU) as our evaluation metric. When comparing with state-of-the-art methods, we adopt Dice coefficient (Dice), Jaccard index (Jac), and Balanced accuracy (Bacc) as our evaluation metrics. The formulas for mIoU, Dice, Jac, and Bacc are as follows:

$$Dice = \frac{2TP}{2TP + FP + FN} \quad (12)$$

$$Jac = \frac{TP}{TP + FP + FN} \quad (13)$$

$$Bacc = \frac{TPR + TNR}{2} \quad (14)$$

$$TPR = \frac{TP}{TP + FP}, FPR = \frac{FP}{FP + TN} \quad (15)$$

$$mIoU = \frac{1}{N} \sum \frac{TP}{TP + FP + FN} \quad (16)$$

4.5. Ablation experiments

4.5.1. Module ablation experiment

To verify the effectiveness of our method, we conduct a substantial number of experiments. In order to better demonstrate the overall performance of our network, we adopt the CAVF task and evaluated it using mIoU as our metric in OCTA_6M. We design a total of nine ablation experiments, as shown in Table 3. Firstly, we explore the performance using three different inputs: OCT, OCTA, and the fusion of OCT and OCTA. We found that the performance of the fusion of OCT and OCTA is weaker compared with using OCTA alone, indicating the presence of modality imbalance problem. Next, we introduced MDFM, multi-stage mid fusion method and MCFM, respectively. The mIoU improved by 0.71% and 0.57% with the addition of MDFM, multi-stage mid fusion and MCFM method, effectively alleviating the modality imbalance problem. Subsequently, we replace the direct summation with VFM during multi-stage fusion, and replace the unidirectional convolution with kernel-adaptive unidirectional convolution. This resulted in an additional improvement of 0.25% and 0.42% in mIoU, respectively. Finally, we use PSM as the 2D backbone of our network. As a result, the mIoU increased by 0.08%. These results collectively demonstrate the effectiveness of our proposed modules.

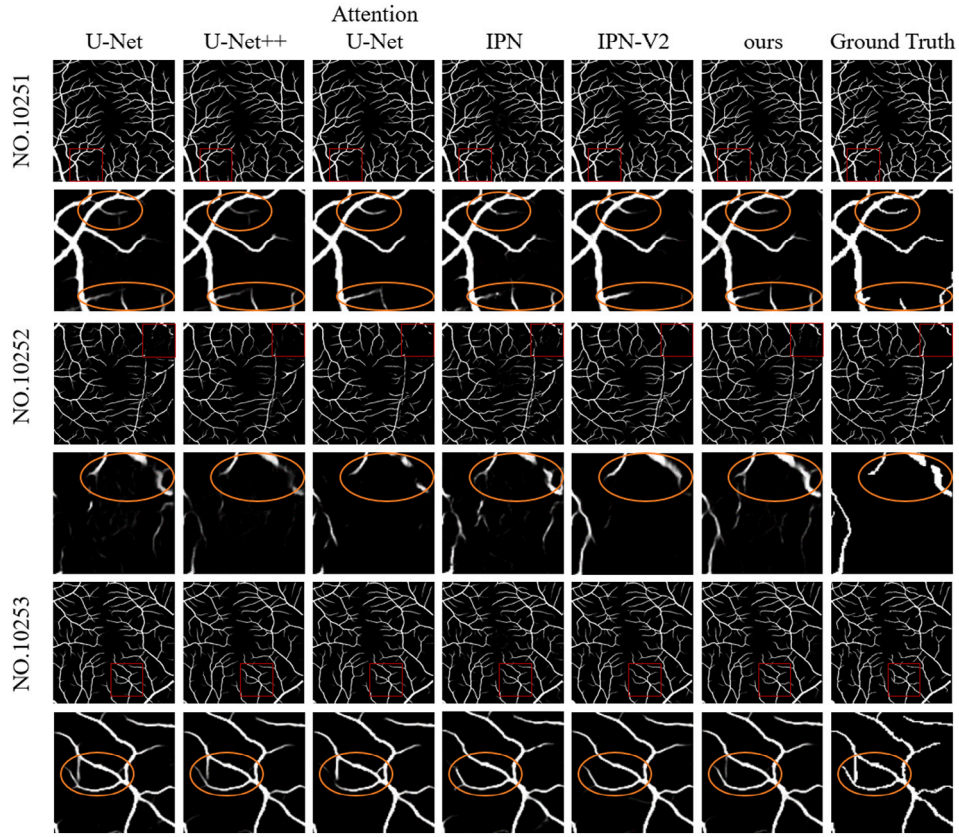


Fig. 10. The segmentation results of Vessel.

Table 3
The impact of different modules on the network.

OCT	OCTA	MDFM	Multi-stage mid fusion	MCFM	VFM	KAPM	2D backbone	IoU				mIoU
								C	A	V	F	
✓							U-Net	32.68	68.36	69.10	70.18	60.08
	✓						U-Net	84.21	76.46	77.12	89.04	81.70
✓	✓						U-Net	81.52	78.49	77.25	87.71	81.24
✓	✓	✓					U-Net	84.09	78.48	76.74	88.50	81.95
✓	✓	✓	✓	✓			U-Net	84.59	79.09	77.61	88.78	82.52
✓	✓	✓	✓	✓	✓		U-Net	84.83	79.31	77.60	89.34	82.77
✓	✓	✓	✓	✓	✓	✓	U-Net	84.95	79.61	78.23	89.59	83.19
✓	✓	✓	✓	✓	✓	✓	PSM	85.10	79.91	78.64	89.44	83.27

Table 4
The impact of network depth on segmentation performance.

3D backbone	2D backbone	Projection times	mIoU
IPN-V2	U-Net	3	81.37
IPN-V2	U-Net	4	81.52
IPN-V2	U-Net	5	81.48

4.5.2. Explore the impact of projection times on segmentation performance

In order to explore the impact of projection times on segmentation performance, we conduct the following experiments as shown in Table 4. In order to achieve more universal results, we adopt IPN-V2 for the projection stage and U-Net for the segmentation stage to perform the CAVF task on the OCTA_{6M} dataset. From the results provided in the table, we can observe that using a deeper network does not necessarily lead to better performance. Many studies have demonstrated similar findings, indicating that deep learning models cannot improve their performance by simply increasing the network depth linearly [42].

4.6. Comparison with state-of-the-art methods

To further verify the performance of our proposed network, we implement RV segmentation and FAZ segmentation in OCTA_{3M} and OCTA_{6M}, and compare them with several state-of-the-art methods. In these methods, The U-Net, Attention U-Net, U-Net++ and Joint-seg 2D segmentation methods use 2D projection images provided by the OCTA-500 dataset. As shown in Tables 5 and 6, our method achieved the best performance on OCTA_{6M} and optimal results in some indicators on OCTA_{3M}. Additionally, OCTA_{6M} contains some patient data, while OCTA_{3M} is data from healthy individuals, indicating that our method has more clinical significance. We visualize some of the segmentation results, as shown in Figs. 10 and 11. Overall, our segmentation results are superior to other methods. For the FAZ segmentation of NO.10252, our method can achieve clearer segmentation. For the vascular segmentation of NO.10251, there are three vessels in the lower orange box, but Attention-U-Net and IPN-V2 are not successful in segmentation. The shape of the vessels segmented in the upper orange box is also closest to Ground Truth. But there are also some over segmentation problems, such as NO.10252 blood vessel segmentation, where our network will segment more vessels.

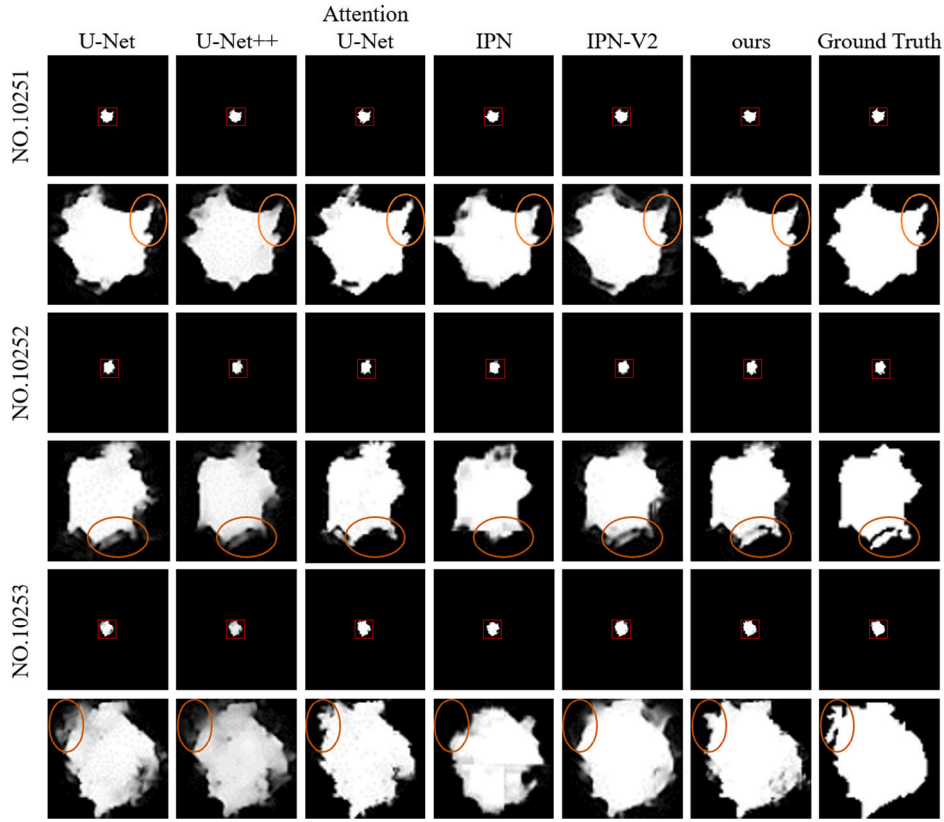


Fig. 11. The segmentation results of FAZ.

Table 5
Comparison with state-of-the-art methods in OCTA_6M.

Method	Vessel			Faz		
	Dice	Jac	Bacc	Dice	Jac	Bacc
U-Net [19]	88.55 $\pm$ 3.43	79.62 $\pm$ 5.08	93.31 $\pm$ 1.68	89.05 $\pm$ 19.19	83.60 $\pm$ 19.70	95.02 $\pm$ 9.69
U-Net++ [21]	88.61 $\pm$ 3.07	79.67 $\pm$ 4.59	91.93 $\pm$ 1.74	88.77 $\pm$ 15.38	82.34 $\pm$ 18.33	93.65 $\pm$ 8.53
Attention U-Net [20]	88.71 $\pm$ 3.19	79.84 $\pm$ 4.74	93.14 $\pm$ 1.49	88.83 $\pm$ 18.89	83.39 $\pm$ 20.89	93.94 $\pm$ 10.25
Joint-seg [43]	89.72 $\pm$ 3.21	81.17 $\pm$ 3.11	–	90.51 $\pm$ 11.13	84.24 $\pm$ 15.44	–
IPN [16]	88.64 $\pm$ 3.21	79.73 $\pm$ 4.92	93.07 $\pm$ 2.42	88.02 $\pm$ 9.91	79.80 $\pm$ 13.81	95.44 $\pm$ 4.63
IPN-V2 [17]	89.08 $\pm$ 2.73	80.41 $\pm$ 4.29	93.52 $\pm$ 2.13	90.68 $\pm$ 8.87	83.97 $\pm$ 12.64	96.42 $\pm$ 4.24
IPN-V2+ [17]	89.41 $\pm$ 2.74	80.95 $\pm$ 4.32	93.46 $\pm$ 2.12	90.84 $\pm$ 8.93	84.23 $\pm$ 12.69	96.37 $\pm$ 4.08
PAENet [25]	89.36 $\pm$ 2.70	80.43 $\pm$ 4.15	94.05 $\pm$ 1.95	–	–	–
PAENet+ [25]	89.69 $\pm$ 2.77	81.42 $\pm$ 4.39	93.68 $\pm$ 2.08	–	–	–
FARGO [44]	89.15 $\pm$ 2.39	80.50 $\pm$ 3.75	–	92.72 $\pm$ 6.74	87.01 $\pm$ 10.60	–
Ours	89.79 $\pm$ 1.95	81.53 $\pm$ 3.07	94.06 $\pm$ 1.35	93.63 $\pm$ 7.85	88.48 $\pm$ 12.22	96.77 $\pm$ 3.55

Table 6
Comparison with state-of-the-art methods in OCTA_3M.

Method	Vessel			Faz		
	Dice	Jac	Bacc	Dice	Jac	Bacc
U-Net [19]	88.55 $\pm$ 3.43	79.62 $\pm$ 5.08	93.31 $\pm$ 1.68	89.05 $\pm$ 19.19	83.60 $\pm$ 19.70	95.02 $\pm$ 9.69
U-Net++ [21]	88.61 $\pm$ 3.07	79.67 $\pm$ 4.59	91.93 $\pm$ 1.74	88.77 $\pm$ 15.38	82.34 $\pm$ 18.33	93.65 $\pm$ 8.53
Attention U-Net [20]	88.71 $\pm$ 3.19	79.84 $\pm$ 4.74	93.14 $\pm$ 1.49	88.83 $\pm$ 18.89	83.39 $\pm$ 20.89	93.94 $\pm$ 10.25
Joint-seg [43]	91.13 $\pm$ 2.09	83.78 $\pm$ 3.40	–	98.43 $\pm$ 0.89	96.93 $\pm$ 1.70	–
IPN [16]	90.62 $\pm$ 5.96	83.25 $\pm$ 7.78	93.87 $\pm$ 4.19	95.05 $\pm$ 4.79	90.91 $\pm$ 7.98	97.68 $\pm$ 3.21
IPN-V2 [17]	92.46 $\pm$ 3.93	86.19 $\pm$ 5.83	95.34 $\pm$ 3.16	97.30 $\pm$ 2.33	94.84 $\pm$ 4.15	98.83 $\pm$ 1.64
IPN-V2+ [17]	92.74 $\pm$ 3.95	86.67 $\pm$ 5.88	95.22 $\pm$ 3.12	97.55 $\pm$ 2.38	95.32 $\pm$ 4.19	98.74 $\pm$ 1.74
FARGO [44]	91.68 $\pm$ 2.05	84.70 $\pm$ 3.34	–	98.43 $\pm$ 0.89	96.84 $\pm$ 1.76	–
Ours	92.83 $\pm$ 2.17	86.67 $\pm$ 3.90	95.25 $\pm$ 2.52	98.41 $\pm$ 1.00	96.89 $\pm$ 1.89	99.14 $\pm$ 1.52

5. Conclusion

In this paper, we propose an Adaptive Projection and Multi-stage Fusion Enhancement Network (APMFENet) for the segmentation of RV, FAZ and CAVF tasks based on two different modal images, namely

OCT and OCTA images. Specifically, we introduce a kernel-adaptive unidirectional convolution for volume projection in multiple stages. At each stage, we employ a multi-stage mid fusion method and replace the simple summation with a Volume Fusion Module to simultaneously capture shallow spatial information and deep semantic information.

Subsequently, we utilize the Multi-stage Complementary Fusion Module to capture the spatial dependencies of different stages in the multi-stage fusion features and obtained complementary features. Meanwhile, for multi-modal features, we use Multi-modal Differential Fusion Module to specialize different modalities and added the fused multi-modal features to the multi-stage features. Finally, the segmentation is performed using a Projection Segmentation Module. Comprehensive experiments demonstrate the excellent performance of our proposed network. In the future research, we plan to improve performance while reducing network parameters and further explore disease classification and prognosis tasks.

CRedit authorship contribution statement

Xiongwen Quan: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Guangyao Hou:** Writing – original draft, Validation, Software, Methodology, Formal analysis. **Wenya Yin:** Formal analysis. **Han Zhang:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgments

This work was supported by Key project of the Natural Science Foundation of Tianjin City [No. 21JCZDJC00140] and the National Natural Science Foundation of China [62373200, 61973174].

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